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Mediating role of eWOM's in green behavior interaction and corporate social responsibility: a stakeholder theory perspective

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Abstract

Purpose – This research aims to explore how policymakers manage the information and communication of green behavior on social platforms to support their growth in corporate social responsibility (CSR). Social platforms play a strategic and interactive role through electronic word-of-mouth (eWOM), which brings unprecedented green purchase opportunities.

Design/methodology/approach – Based on stakeholder theory, a conceptual framework is designed to investigate the influence of green behavior interactions (GBIs) on CSR under the mediating effects of eWOM subfactors (eWC = eWOM communication, eWIA = eWOM information adoption and eWSC = eWOM source credibility). Data from 414 regular stakeholders of logistics firms were analyzed via structural equation modeling.

Findings – The results reveal positive influences of the GBI on eWC, eWIA, eWSC and CSR, with path coefficients of 0.329, 0.713, 0.809 and 0.316, respectively. The mediating effects of eWC and eWSC from the GBI to CSR were discovered with path coefficients of 0.105 and 0.226, respectively. Coincidentally, the mediating role of eWIA was positive but not supported. The outcomes of this study indicate that the administration of GBI and eWOM from a green purchase perspective is essential for a firm. The CSR practices of green logistics firms can be successfully supported by the administration of the GBI and eWOM indicators.

Originality/value – This study develops a novel multidimensional framework that illustrates the impact of eWOM on reducing information asymmetry, enhancing credibility, supporting informed decision-making and improving green consumer behavior. By amplifying positive reviews, increasing engagement and establishing a feedback loop, this framework aims to provide comprehensive insights into the efficacy of eWOM for firms' products and services.

Keywords Corporate social responsibility, Green behavior, eWOM, Information credibility, Stakeholder theory perspective

Paper type Research paper

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1. Introduction

In the realm of green business paradigms, corporate social responsibility (CSR) is defined as a firm's ability to conduct operations in a manner conducive to the well-being of the economy, societal welfare and environmental sustainability (Ahmad *et al.*, 2024; Chu *et al.*, 2023). According to Caulfield and Lynn (2024) CSR refers to a company's commitment to operate in a socially, environmentally and economically sustainable manner while being accountable to its stakeholders, including employees, customers, investors, communities and the environment. CSR is a vital part of corporate governance, and corporations create a positive impact on society and the environment (Caulfield and Lynn, 2024; Qin and Luo, 2021; Rendtorff, 2020). China's rapid economic development has led to environmental and social challenges, pressuring companies to adopt CSR practices. CSR initiatives often target pollution reduction, labor rights and community engagement, and stakeholders have asked firms to be more liable for such initiatives (Liu *et al.*, 2023). As the middle class grows, consumers demand higher ethical standards from companies, and CSR becomes a differentiator. The current growing awareness of CSR has led firms to promote social initiatives and green behavior through digital platforms (Jiang and Park, 2022; Park and Jiang, 2020). Expert content analysis shows that digital platform reviews and comments are crucial to green CSR evaluation (Xu *et al.*, 2023; Knight *et al.*, 2022). Social intelligence from apps and networks is priceless. The influence of consumer attitudes strengthens the green environment connection between consumers and firms through sharing information on social sites. (El Ghoul *et al.*, 2019), which could stimulate favorable feedback (Fernández *et al.*, 2022; Shokoohyar *et al.*, 2023). Today, digital platforms have given great means and probabilities to share online feedback about products (Liao *et al.*, 2023). Virtual community forums such as media websites, Twitter, Facebook, WeChat and blogs are popular digital platforms for sharing feedback, information and knowledge (El Ghoul *et al.*, 2019; López Jiménez *et al.*, 2021).

Digital social sites (such as Facebook, Instagram, Twitter, LinkedIn and others) play a crucial role in CSR by providing a platform for companies to engage, communicate and amplify their CSR efforts (Jacobson *et al.*, 2020). Chu and Chen (2019) argued that digital platforms provide new ways for firms to express and inform customers about their activities and achievements in a green environment. CSR is affected by social platforms with high communication efficiency due to the high transmission of online feedback (Etter, 2013; Jacobson *et al.*, 2020). Businesses are adopting innovative techniques to improve CSR to address the green environment and green behavior (Etter, 2013; Fox and Warber, 2015; Sherrick and Hoewe, 2018), and the present research examines logistics enterprises' CSR initiatives with diverse information-based conceptions. In China, businesses and consumers are connected through online buying and selling platforms, most of which are social platforms. China's rapid digital development and distinctive sociopolitical framework can teach other countries, especially emerging markets. Current research explores how businesses can use social media to enhance CSR through word-of-mouth (WOM) from policymakers and customers. Chinese stakeholders are more likely to connect with CSR projects that promote social harmony and communal values due to China's collectivist culture. Academic research has explored the relationships between CSR, green technology adoption and business performance, confirming its importance in modern corporate dynamics.

WOM is the informal sharing of feedback, views and recommendations regarding products, services, brands and companies (Mukhopadhyay *et al.*, 2023). Kim and Yoo (2020) argued that WOM is associated with discussing product quality and brand/firm awareness by consumers. It is the oldest technique for trustworthy and credible communication with friends, family, coworkers or other trusted individuals, making it powerful (Baker *et al.*, 2016). WOM used to be a better way to interact with users and firms to improve satisfaction, efficiency and loyalty than direct services did (Chawdhary and Dall'Olmo Riley, 2015; De Angelis *et al.*, 2012). Today, with the advent of the internet, the literature now uses electronic word-of-mouth (eWOM) to describe WOM (Donthu *et al.*, 2021), which has grown in prominence in influencing public opinion and green purchasing decisions. Fernández *et al.* (2022) explained that, compared with personal communication, online communication is more convenient, long-lasting and quantifiable and involves multiway information sharing. In the industrial sector, the importance of eWOM communication (eWC) in modern companies is considerable for several developments and helps to increase firm value, the business innovation model, knowledge management and social responsibility (Fernández *et al.*, 2022; Markovic *et al.*, 2022; Sherrick and Hoewe, 2018). When communicating via eWOM, firms feel free to engage with their stakeholders and conversate about CSR activities and green behavior (Verma *et al.*, 2023). China's unique cultural, political and economic environment necessitates a localized analysis of CSR and eWOM (Ahmed *et al.*, 2020; Chu and Chen, 2019; Chu *et al.*, 2020). By deeply integrating local cultural, political and economic factors into its analysis, current work could significantly enhance the firm's position in social responsibilities. The link between CSR and eWOM offers significant scholarly and practical potential, especially when addressed in the context of China. Because, by strategically aligning CSR initiatives with cultural, regulatory and socioeconomic factors, companies in China can harness the power of eWOM to bolster their reputation, foster trust and achieve long-term success. However, this study under discussion would be significantly stronger if it more deeply connected its findings to China's specific conditions and addressed green behavior interactions (GBIs).

Basically, green behavior, or actions motivated by environmental consciousness, is becoming an integral aspect of CSR and consumer decision-making worldwide (Eberle *et al.*, 2013; Shen, 2021; P. Zhang and Benjamin, 2007). In China, green behavior is influenced by a blend of traditional beliefs, government initiatives and increased consumer awareness (Schaltegger *et al.*, 2019). For example, Confucian ideas emphasizing harmony with nature fit with modern sustainability goals, establishing a cultural basis that promotes green CSR initiatives. The role of GBIs in shaping CSR is an area of growing interest as stakeholders seek to understand how these interactions can impact a company's green behavior and CSR practices. The GBIs concept refers to exchanging information between a company and its stakeholders for changes in the green behavior of a product (Eberle *et al.*, 2013; Shen, 2021; P. Zhang and Benjamin, 2007). GBIs typically happen between individuals, between consumers and businesses, or within communities, focusing on promoting and adopting eco-friendly and sustainable practices. These information interactions can take many forms, including corporate reporting, stakeholder engagement and disclosure (Topal *et al.*, 2020). GBIs are essential for fostering a culture of sustainability and influencing consumer behavior, which is under investigation. Nevertheless, the impact of these GBIs on CSR is complex and multifaceted and needs to be investigated. In the digital era, enterprises can both disseminate and procure data pertaining to their CSR endeavors, milestones and best practices via online channels (Liu *et al.*, 2024b; Veile *et al.*, 2022). Reducing carbon emissions, optimizing fuel consumption or adopting green technologies are considered the most crucial challenges for Chinese logistics companies. Addressing CSR challenges in logistics not only improves environmental sustainability but also enhances operational efficiency, builds brand loyalty and meets increasing consumer demand for responsible practices (Rendtorff and Bonnafous-Boucher, 2023). Logistics companies in China are proactively addressing CSR challenges by implementing various strategies aimed at a green environment. For example, logistical firms can optimize all transportation routes and consolidate the shipments to minimize the emissions for customer satisfaction through gaining information from digital platforms.

Moreover, the credibility of these information sources is paramount, as stakeholders demand assurance that the information is both dependable and significant. eWOM source credibility (eWSC) is crucial for CSR because shared information can reach many viewers quickly (Hoozée *et al.*, 2019), and online comments and opinions about a company's CSR initiatives can affect customers, investors, employees and the public (Liu *et al.*, 2024a). During CSR-related disasters or controversies, eWOM can either amplify or mitigate damage. Credible sources can provide accurate information and context to manage CSR-related disasters. Credible eWOM sources can assist organizations in enhancing their CSR strategy with constructive criticism and ideas. Next, for firms, eWOM information adoption (eWIA) is key to success, which may help improve their green behavior and CSR image. For example, confidence increases when influencers, experts and renowned online groups contribute eWOM knowledge. If information adoption is convincing, consumers can alter their preferences on the basis of several preconditions, such as the credibility of information, usefulness and adoption of information (Verma *et al.*, 2023; Vollero *et al.*, 2019). Based on the above discussion, the eWSC, eWIA and eWC concepts are being investigated as three subelements of eWOM. Hence, the current work explores the potential roles of GBIs and eWOM elements to contribute to the literature and two research questions are posed:

RQ1. How do green behavior interactions impact electronic word-of-mouth and corporate social responsibility?

RQ2. Does electronic word-of-mouth play a mediating role among green behavior interactions and corporate social responsibility?

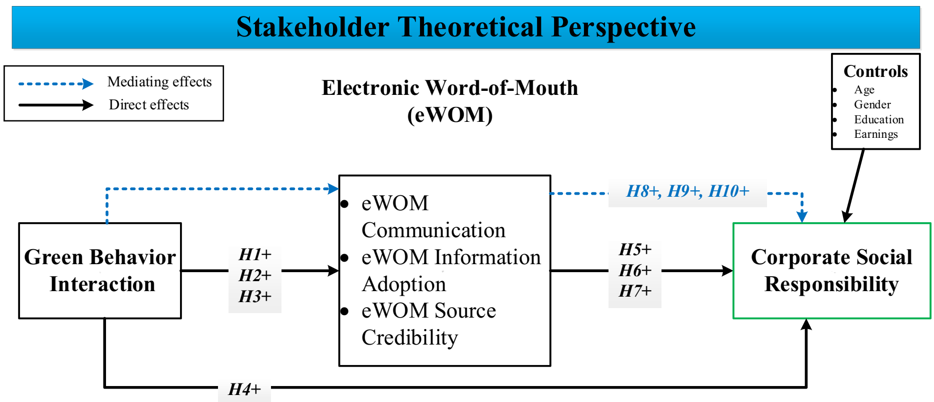
The structure of this article aligns with the research objectives set forth. Section 2 delves into the theoretical underpinnings that inform our conceptual framework. In Section 3, we articulate the hypothesis development. Section 4 outlines the methodology, detailing data sourcing, the analytical approach, empirical mechanisms and hypothesis verification. Section 5 offers a discussion of the empirical findings along with the implications. Finally, Section 6 presents the conclusions, accompanied by implications and limitations of the research.

2. Literature review

2.1 Stakeholder theory (theoretical perspective)

The theoretical foundation of this study draws from a seminal model for stakeholder theory (ST). According to Odunjo *et al.* (2023), ST suggests that businesses exist not only to generate profits for shareholders but also to create value for all stakeholders who are impacted by or can impact the organization's operations. Where, business aim to generate benefits for balancing for all collaborators to maximize the collective benefits (Freeman, 1984) and reports morals in the administration of organizational issues (Rendtorff and Bonnafous-Boucher, 2023). The origin of STs can be traced to the 1960s, which was proposed by the Standford Research Institute as the concept of stakeholders (Parmar *et al.*, 2010). ST is a perspective on capitalism that highlights the interdependencies between a firm and its stakeholders by perceiving green behavior, which leads to active communications, information adaptability and credibility (Bonnafous-Boucher and Rendtorff, 2016; Liu *et al.*, 2023; Shokoohyar *et al.*, 2023). Freeman and David (1983) explained that individuals or groups who can affect or be affected by an organization's activities, policies or goals are stakeholders, which emphasizes an organization's responsibility to create value for its stakeholders by addressing their needs and expectations. With respect to stakeholder's expectations, a company should consider the interests of all stakeholders (employees, customers, communities and even the environment) in its decision-making processes (Adomako and Dong Tran, 2022). ST is crucial for achieving competitive goals and holistic business success in an interconnected world.

Where CSR refers to a company's efforts to address societal and environmental concerns beyond financial objectives (Adomako and Dong Tran, 2022). ST provides the framework for integrating CSR into logistics by focusing on the interests of all affected parties. For example, logistics systems affect numerous stakeholders, including suppliers, customers, employees and local communities. ST ensures that these impacts are considered, leading to ethical supply chain practices (Markovic *et al.*, 2022; Shokoohyar *et al.*, 2023). Logistics often contributes to carbon emissions and environmental degradation. ST aligns CSR goals with stakeholder expectations, driving initiatives like green logistics, energy efficiency, waste reduction (Odunjo *et al.*, 2023), eWOM communication, information adoption and source credibility act as mechanisms through which stakeholders evaluate and reinforce perceptions of an organization's CSR efforts (Adomako and Dong Tran, 2022). Schaltegger *et al.* (2019) argued that the major role of ST is to derive core intentions from the positive or negative evolution of green conduct, whereas subjective norms, attitudes and actions from individuals' green behavior. ST emphasizes that organizations must consider the interests of all stakeholders (e.g. consumers, employees, communities and regulators) to achieve long-term success (Parmar *et al.*, 2010). However, in Figure 1, hypotheses related to GBIs (e.g. *H1*, *H2* and *H3*) highlight how stakeholders' eco-friendly actions influence eWOM (communication, adoption and credibility). It assumes that green behavior aligns with stakeholder expectations, which fosters engagement in eWOM. Next, hypotheses *H5–H7* predict direct links between eWOM components (e.g. adoption and credibility) and CSR performance. Where ST posits that transparent and credible communication builds trust. eWOM acts as a mechanism for stakeholders to assess a company's CSR. For mediating role of eWOM, ST theory recognizes the role of communication channels in strengthening



Note: where "H" denotes the proposed hypothesis
Source: Figure by authors

Figure 1. Conceptual framework

relationships between stakeholder actions (green behavior) and organizational outcomes (CSR). The mediation hypotheses imply that eWOM amplifies stakeholder influence on CSR. Sahi *et al.* (2022) explained that if a company wants to maintain its legitimacy for a longer period of survival, it needs to ensure that the meaningful eWOM dynamics are followed by stakeholders. Customers are more inclined to support and engage with a company that demonstrates a commitment to environmentally sustainable practices ("green behavior") and CSR. The involvement of customers in CSR contributes to a firm's positive public image and improves the company's profile in the eyes of stakeholders and customers (Odunjo *et al.*, 2023; Sahi *et al.*, 2022). A firm's green behavior activity can establish consumer faith in the brand, which may lead to increased attendance (Chu *et al.*, 2020). The current research successfully used ST to formulate a hypothetical research model to explain the relationships among stakeholders' green behaviors in different contexts, such as GBI, eWOM communication, eWOM information adoption, eWOM source credibility, green behavior and CSR (see Figure 1). ST argues that individuals tend to identify with a product or a firm with their emotional attachment and identification. In the literature, researchers have explained how companies are engaged in CSR practices, which can be affected by consumers' sense of belonging (Adomako and Dong Tran, 2022; Gallego-Álvarez and Pucheta-Martínez, 2022).

2.2 Hypothesis development

Traditionally, CSR discussion has focused on improving stakeholder welfare for competitive reputational advantages with the public or consumers (Park and Jiang, 2020), which addresses green behavior that avoids unethical social issues (Chou *et al.*, 2023; Ikram *et al.*, 2020). Firms face criticism of their harmful behavior toward the environment, pollution and unfair treatment of employees and suppliers (Ahmad *et al.*, 2024; Chu *et al.*, 2020). During the criticism period, shared information on online platforms played a significant role in resolving social issues. Moreover, the public's irresponsible attitude toward the green environment decreases employees' motivation, damages reputation and causes consumer losses, which depend on WOM communications (Fernández *et al.*, 2022). Previous studies have shown a significant disparity in the cognitive processes that people experience before experiencing emotions. In this case, interactions from sharing information and adopting information among individuals, organizations or communities are examined, which refers to exchanging information within an organizational system. Interactive behavior plays a crucial role in facilitating knowledge exchange, collaboration and innovation. Therefore, effective GBIs within the organizational system can enhance entrepreneurial activities, value creation and the system's overall functioning (Vollero *et al.*, 2019).

Various studies have examined the effect of GBI on information from online feedback, including customers' purchase intentions, destination visit intents and brand love. For example, Abubakar *et al.* (2017) reported that eWOM activates consumer green behavior postpurchase and reduces negativity biases among family and friends following a pleasant service contact. To drive the generation of eWOM, several motivations, such as prepurchase expectations, satisfaction or dissatisfaction, consumer delight and general behavior, have been identified in the literature. In addition, young consumers' green behavior has been investigated through the eWOM effect, and the finding shows that eWOM influences as well as engaged youth (Zhang *et al.*, 2017). eWOM significantly influences consumer perceptions and behavior, particularly with respect to green behavior and sustainability. Customers typically use eWOM to validate their purchase decisions wherever they are. A large number of favorable online discussions about a company's green activities (e.g. energy-efficient goods and trash reduction programs) push others to choose environmentally friendly solutions. In eWOM communication, the effects of visual information such as videos and pictures are highly significant in terms of credibility, product interest, purchase intention and message quality compared with those without visual information, such as articles, newspapers and books (Lin *et al.*, 2012). The role of information interaction in product interest and purchase intentions is revealed among product categories and visual information (Bi *et al.*, 2019). User-generated content in eWOM platforms (blogs, forums, social media) educates consumers about green products and practices. The influence of eWOM interaction has a positive effect on customers' purchase intentions, even consumers who share their green experiences (e.g. posting about using reusable packaging or supporting brands that reduce emissions) inspire others to engage in similar behaviors (Yusuf *et al.*, 2018). Rossmann *et al.* (2016) highlighted the significant impact of positive emotional experiences on eWOM generation and highlighted the importance of eWOM in the digital era. Le *et al.* (2023) examined eWOM processing from the receiver's perspective, conceptualized the connections and highlighted the significance of eWOM for consumers because they shift from offline to online platforms. Moreover, the effect of CSR conversation through WOM intentions on a firm's reputation has been investigated (Fatma *et al.*, 2020). Eberle *et al.* (2013) suggested that using interactive channels to communicate about CSR can have a positive effect on corporate reputation and eWOM, with higher message credibility and stronger feelings of identification with the firm through perceived interactivity in CSR. From the above discussion, the impact of GBI on eWOM communication, eWOM information adoption, eWOM source creditability and CSR is assumed in four proposed hypothesized relationships, which are given below:

H1. GBI has a positive influence on eWOM communication.

H2. GBI has a positive influence on eWOM information adoption.

H3. GBI has a positive influence on eWOM source creditability.

H4. GBI has a positive influence on eWOM corporate social responsibility.

CSR initiatives are positively linked with consumers' eWOM conversations (Fatma *et al.*, 2020). Digital users leverage social media websites and apps to produce green images that influence human green behavior (Liu *et al.*, 2024b). Online reviews that provide customers' feedback about goods, services and firms are examples of eWOM communication; those reviews are a reliable source of information untainted by consumers and producers (Verma *et al.*, 2023). The relationships between green initiatives and eWOM have been investigated in different studies (Chu *et al.*, 2020; Hoozée *et al.*, 2019; Vollero *et al.*, 2019). In the literature, connections between eWOM and CSR have been explored with the goal of a greener environment. For example, the influence of CSR on WOM is positive and moderated by price perception, which provides a better understanding of the role of CSR in generating WOM (Han *et al.*, 2020). Shen (2021) developed a persuasive eWOM model by focusing on the relationships between influencers' eWOM messages and consumer participation on social media. The impact of CSR on brand development was investigated for CSR initiatives through online platforms. In this research, we investigate the mediating role of eWOM communication, eWOM information adoption and eWOM source creditability for CSR valuation through the direct effect of GBI. Iankova *et al.* (2019) reported that eWOM has gained significant importance in the digital age because of the widespread use of social media platforms and online review communication. eWOM has become a powerful tool for end users to gather information, seek recommendations and make purchase decisions through communication (Gharib *et al.*, 2020). Furthermore, eWOM allows individuals to share their opinions and experiences with a potentially large audience, influencing the perceptions and green behaviors of others (Herhausen *et al.*, 2019). Online communication between two parties involves changing green behavior by providing benefits from one party's feedback to another (Etter, 2013). In brief, eWOM communication about social features or information online platforms could also be beneficial for enhancing CSR. eWOM communication is considered an activity of information transformation among people who receive and provide essential information (Fatma *et al.*, 2020). High participation in eWOM activities means that the source of CSR communication might affect customer attitudes and actions toward a green environment (Chu and Chen, 2019). Next, eWOM information adoption in the context of eWOM refers to the process by which individuals accept and use the information shared through eWOM channels in their decision-making and subsequent green behaviors. The adoption of eWOM information is also influenced by consumers' motives to engage with eWOM and eWOM information adoption by end users could occur for various reasons, such as information validation, product evaluation and postpurchase validation (Wang *et al.*, 2021). The quantity and quality of eWOM, as well as consumers' attitudes toward eWOM, play a critical role in the evaluation of information usefulness and adoption. The effect of the provided information on human green behavior could be speckled from one person to another person because the same message can elicit the receivers' thoughts (Ghezal, 2024; Liu *et al.*, 2024b). In the literature, scholars have examined the information exchange process to understand how people assume a satisfactory level for adopting information sought from eWOM (Donthu *et al.*, 2021). Furthermore, several factors influence the adoption of eWOM information. Attitudes toward eWOM play a significant role in how consumers engage with and adopt eWOM content (Aghakhani *et al.*, 2018). Kaur and Kaur (2023) explained that positive attitudes toward eWOM are associated with greater interest in information, regardless of its credibility. Consumers develop attitudes toward eWOM on the basis of their perception of its credibility, including factors such as appearance, presentation, transparency and the identity of the source (Ismagilova *et al.*, 2020; Mahapatra and Mishra, 2017). The credibility of eWOM information is judged on the basis of the trustworthiness and expertise of the individuals generating the content. The quality and credibility of eWOM also impact information adoption. Consumers evaluate the usefulness of eWOM information on the basis of its quality and credibility (Teng *et al.*, 2014). The more useful and credible the information is perceived to be, the greater the likelihood of adoption. In addition, the quantity of eWOM, such as the number of reviews or recommendations, can influence consumers' evaluation of information usefulness and adoption. On the basis of the above discussion, the direct and mediating effects of eWOM communication, eWOM information adoption and eWOM source credibility are proposed:

H5. eWOM communication has a positive influence on CSR.

H6. eWOM information adoption has a positive influence on CSR.

H7. eWOM source credibility has a positive influence on CSR.

H8. eWOM communication has a positive mediating influence on the relationship between CSR and GBI.

H9. eWOM information adoption has a positive mediating influence on the relationship between CSR and GBI.

H10. eWOM source credibility has a positive mediating influence on the relationship between CSR and GBI.

3. Methodology

3.1 Sample and data source

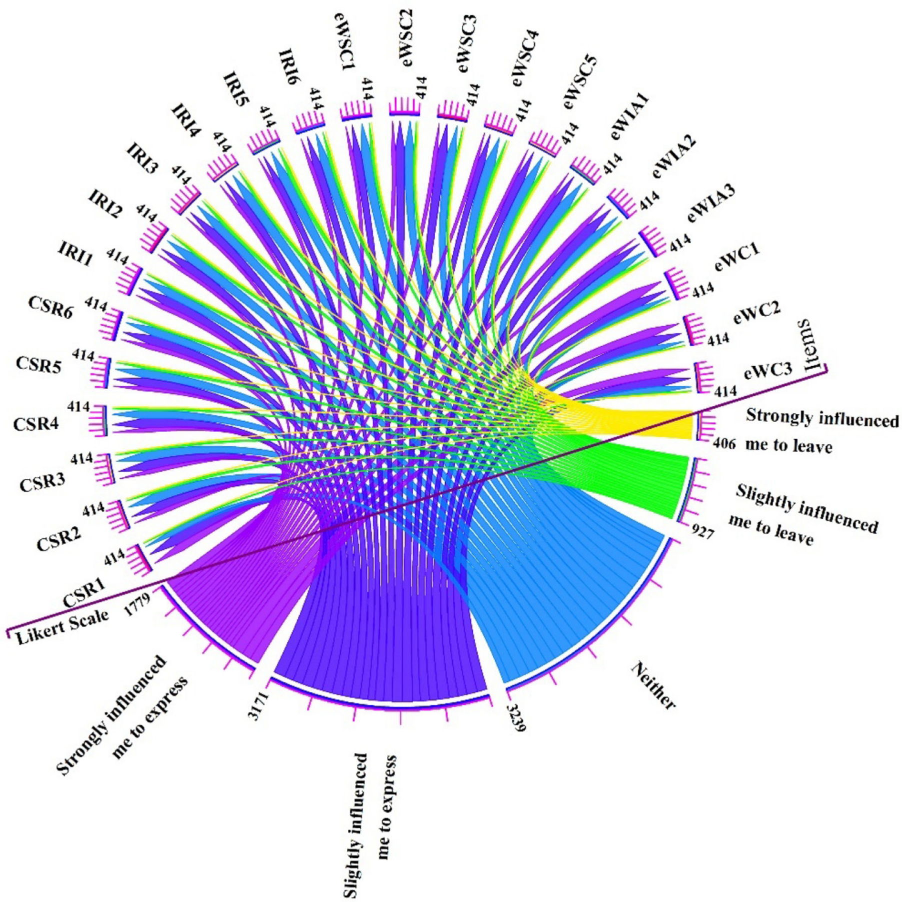
The data used in the analysis were gathered from a survey conducted on logistics firms (i.e. Tiantian Express, China Railway Express, Jingdong*, Zhongtong*, Yunda*, Shentong, Youzheng*, Shunfeng Express *, Jitu, Dangde, Yibang Express and Xingchen Express) in China. The massive amount of online shopping and selling on various platforms, including Taobao, Jing Dong, Meituan, Weipinghui, E Le Ma, Dazhongdianping and Koubei, is the primary rationale behind our selection of these logistics providers. In recent years, because of the development of logistical networks and online platforms, the Chinese market for online business has been relatively wholly integrated and well-developed. For data collection, a list of logistics companies was generated by scrutinizing the Chinese national data center Tian Yan Cha. A well-designed questionnaire was used as a data collection instrument for the survey. The respondents were identified as regular users of logistical services involving online shopping or platforms. To validate the instrument, a group of five experts related to CSR, green products and the environment was arranged. This group verified the validity of the data collection instrument after the excellence of all the questions was analyzed. A minor modification was suggested by the experts in wording the scale items and the sentence structure, which were incorporated to assess validity. We used a pilot test for clarity and validity assessment of the final questionnaire, which was directed at 15 respondents from the target population. The pilot test confirmed the validity and clarity of the final questionnaire.

The survey questionnaire was distributed to collect the data, measure all included constructs and assess the structural model. The data collection instrument included ordinal scale questions (e.g. gender, age, income, education, experience and marital status) in the first section, and in the second section, adapted scale items for construct measurement were included. In addition, each scale item was rated by respondents on a five-point Likert scale ranging from strongly influenced me to express (5), slightly influenced me to express (4), neither (3), slightly influenced me to leave (2) and strongly influenced me to leave (1). In addition, the final data set for 414 responses was visualized through a chord diagram graphical method and offered follow-up connections between the Likert scale and 23 items (see Figure 2). According to the visualization results, most of the respondents expressed strong and slight feelings in favor of all the questions during the survey.

Among the 414 respondents, 372 were owners and 42 were working in logistical firms. With respect to education level, 176 respondents had bachelor’s degrees, 236 had master’s degrees, 1 respondent had a PhD degree and the other option was selected for the remaining 1 respondent. The age classification and explanation of the respondents were 29 = <25 years, 39 = 25–30 years, 140 = 30–35 years, 179 = 35–40 years and 27 = above 40 years. Among the 414 respondents, 363 were men and 51 were women, as presented in Table 1. The earning classification and explanation of respondents for one year are 34 = RMB <200k, 129 = RMB 201k–250k, 224 = RMB 251k–300k and 27 = earnings above RMB 300k.

3.2 Data analysis

The data analysis was executed through SmartPLS 4.0.9.2 statistical software. In the analysis process, factor analysis was used in the first stage. To prove a normal univariate distribution of the data, excess kurtosis and skewness values were examined carefully. The calculated excess kurtosis and skewness values were within the limit (see Table 1), the standardized limit for skewness was between −2 and +2 and kurtosis was between −7 and +7 (Hair *et al.*, 2020). Variance inflation factor (VIF) values were calculated for each item that was less than 5 (see Table 2). A VIF value is a measure of the amount of multicollinearity, which should be



Notes: where eWC = eWOM communication; eWIA = eWOM information adoption; eWSC = eWOM source credibility; CSR = corporate social responsibility; GBI = green behavior interactions

Source: Figure by authors

Figure 2. Visualization of the responses

less than 5 because a value less than 5 indicates a low correlation among predictors (Doty and Glick, 1998; Hair *et al.*, 2020).

Furthermore, a structural model was created, and the construct reliability and validity were considered to examine the model assessment. The model assessment was based on convergent validity (CV) and divergent validity (DV) measurements, which were restrained by factor loading and average variance extraction (AVE) values. For reliability and validity of the constructs, AVEs, composite reliability (CR) and Cronbach's alpha (CA) values were calculated by using factor loading values of scale items (Hair *et al.*, 2020). The AVEs were

Table 1. Demographical information of participants

Demographical constructs	Constructs	Frequency	%
Gender	Male	*363*	87.7
	Female	*51*	12.3
Age	<25 years	*29*	7.0
	25–30 years	*39*	9.4
	30–35 years	*140*	33.8
	35–40 years	*179*	43.2
	>40 years	*27*	6.5
Education level	Bachelor's degree	*176*	42.5
	Master's degree	*236*	57.0
	Doctorate	*1*	0.2
	Other	*1*	0.2
Annual earnings	RMB 200k	*34*	8.2
	RMB 201k–250k	*129*	31.2
	RMB 251k–300k	*224*	54.1
	RMB 301k–350k	*27*	6.5
Total		*414*	*100*

Source: Created by authors

Table 2. Construct reliability and validity

Constructs	Items	Factor loading	VIF	AVEs	Cronbach's alpha	Excess kurtosis	Skewness
Corporate social responsibility	CSR1	0.659	1.461	0.705	0.914	0.455	−0.823
	CSR2	0.888	3.559			−0.214	−0.283
	CSR3	0.873	3.384			−0.059	−0.341
	CSR4	0.894	3.487			0.124	−0.520
	CSR5	0.829	2.364			−0.176	−0.347
	CSR6	0.871	2.948			−0.095	−0.433
Green behavior interaction	GBI1	0.886	3.505	0.782	0.944	−0.250	−0.300
	GBI2	0.899	3.750			−0.105	−0.353
	GBI3	0.888	3.326			−0.346	−0.277
	GBI4	0.883	3.363			−0.274	−0.292
	GBI5	0.843	2.689			−0.522	−0.120
	GBI6	0.903	3.923			−0.395	−0.304
eWOM communication	eWC1	0.875	2.077	0.807	0.880	0.384	−0.988
	eWC2	0.913	2.894			0.375	−0.820
	eWC3	0.906	2.727			0.207	−0.791
eWOM information adoption	eWIA1	0.872	2.027	0.714	0.800	0.010	−0.210
	eWIA2	0.833	1.806			0.118	−0.503
	eWIA3	0.831	1.542			−0.378	−0.297
eWOM source credibility	eWSC1	0.861	2.604	0.713	0.899	−0.293	−0.228
	eWSC2	0.867	2.725			−0.067	−0.385
	eWSC3	0.853	2.418			0.070	−0.452
	eWSC4	0.820	2.092			−0.306	−0.159
	eWSC5	0.820	2.089			−0.211	−0.343

Notes: eWC = eWOM communication; eWIA = eWOM information adoption; eWSC = eWOM source credibility; CSR = corporate social responsibility and VIF = variance inflation factor
Source: Created by authors

above 0.5 (threshold), which reflects the convergent validity of the constructs (see Table 2). For DV examination, the Fornell–Larcker criterion, which is a method used to examine the DV, was used. In this method, the square root of the AVEs should be greater than the correlations (Fornell *et al.*, 1981). The results show that the correlation values are lower than the square root of the AVEs, which demonstrates that the constructs are valid on the basis of the DV (see Table 3). In addition, the age, gender, education and earnings of the respondents were controlled for in terms of CSR during the structural equation modeling (SEM) analysis. The results for the control variables were insignificant for the structural model.

3.3 Empirical mechanism

For the empirical mechanism, this study builds stepwise regression lines to examine the proposed hypothesis for direct and mediating effects following (Chen *et al.*, 2022). Current research uses CSR as an explanatory construct, GBI as a core explanatory construct and eWOM indicators (eWC, eWIA and eWSC) as intermediary constructs.
Regression lines for direct effects:

$$eWC_{it} = a_0 + a_1 IRI_{it} + a_5 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (1)$$

$$eWIA_{it} = c_0 + c_1 IRI_{it} + c_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (2)$$

$$eWSC_{it} = d_0 + d_1 IRI_{it} + d_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (3)$$

$$CSR_{it} = h_0 + \; + h_1 IRI_{it} + h_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (4)$$

$$CSR_{it} = i_0 + \; + i_1 eWC_{it} + i_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (5)$$

$$CSR_{it} = d_0 + \; + d_1 eWIA_{it} + d_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (6)$$

$$CSR_{it} = t_0 + \; + t_1 eWSC_{it} + t_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (7)$$

Regression lines for intermediary effects:

$$Med(eWC_{it}) = r_0 + r_1 IRI_{it} + r_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (8)$$

Table 3. Discriminant validity

Discriminant validity (Fornell–Larcker criterion)	CSR	GBI	eWC	eWIA	eWSC
Corporate social responsibility	*0.840*				
Green behavior interaction	0.732	*0.884*			
eWOM communication	0.577	0.329	*0.898*		
eWOM information adoption	0.679	0.713	0.401	*0.845*	
eWOM source credibility	0.746	0.809	0.381	0.746	*0.844*

Source: Created by authors

$$CSR_{it} = f_0 + f_1 IRI_{it} + f_2 Med(eWC_{it}) + f_3 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (8a)$$

$$Med(eWIA_{it}) = b_0 + b_1 IRI_{it} + b_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (9)$$

$$CSR_{it} = g_0 + g_1 IRI_{it} + g_2 Med(eWIA_{it}) + g_3 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (9a)$$

$$Med(eWSC_{it}) = y_0 + y_1 IRI_{it} + y_2 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (10)$$

$$CSR_{it} = l_0 + l_1 IRI_{it} + l_2 Med(eWSC_{it}) + l_3 controls_{it} + age + gender + education + earning + \acute{\varepsilon} \quad (10a)$$

where equations (1)–(4) represent the empirical mechanism for classifying the direct influence of GBI on eWOM indicators and CSR. Equations (5)–(7) present the empirical mechanism used to identify the direct impacts of eWC, eWIA and eWSC on CSR individually. Finally, the intermediary effects of eWOM indicators are identified in combined equations, i.e. the mediating role of eWC (*MedeWC*) is combined in (8), the mediating role of eWIA (*MedeWIA*) is combined in (9) and the mediating role of eWSC (*MedeWSC*) is combined in (10). Furthermore, in the equations, the castoff of the subscripts *i and t* is the classification of the firm and year.

3.4 Structural equation modeling results (hypothesis testing)

A bootstrapping technique known as a nonparametric technique was used to test the statistical significance of the proposed hypothesis in the PLS-SEM results. Bootstrapping is a resource used to create a subsampling list by randomly choosing observations from the original data. Then, by using the subsample data, the path model for PLS is estimated with a 5,000 bootstrapping resample algorithm. The model fitness was computed through the standardized root mean square residual (SRMR), the squared Euclidean distance (d_ULS), the geodesic distance (d_G), Chi-square and normed fit index (NFI) values. The computed values of the model fit indices were acceptable as suggested criteria (see Table 4). Furthermore, path coefficients, R-squared values and levels of significance with t-values were computed for the examination of hypothetical relationships. Among the total proposed hypotheses, nine were significantly supported and two were insignificant.

Table 4. Model fit index

Fit indices	Saturated model	Estimated model
Standardized root mean square residual (SRMR)	0.061	0.075
Squared Euclidean distance (d_ULS)	1.423	2.105
Geodesic distance (d_G)	0.613	0.675
Chi-square	1,394.287	1,468.291
Normed fit index (NFI)	0.839	0.83

Source: Created by authors

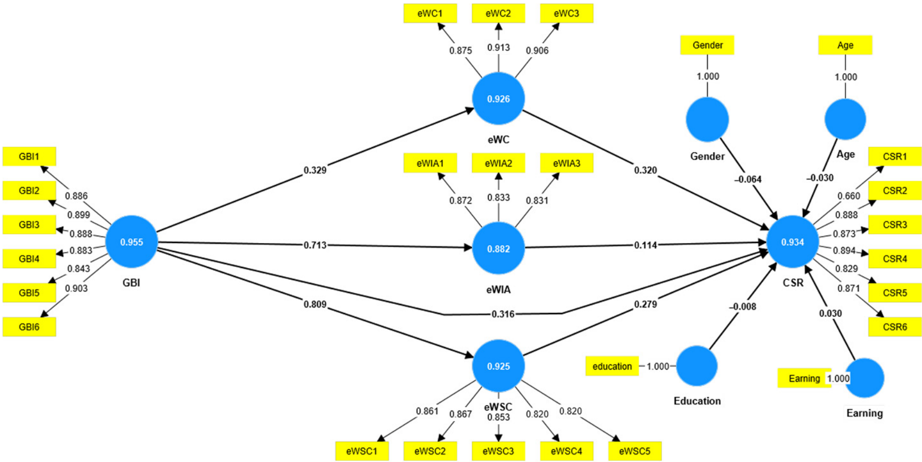
The results show that the GBI has a significant effect on three elements of eWOM, i.e. eWC (*H1*: $\beta = 0.329$, $t = 6.407$ and $p < 0.000$), eWIA (*H2*: $\beta = 0.713$, $t = 21.775$ and $p < 0.000$) and eWSC (*H3*: $\beta = 0.809$, $t = 29.370$ and $p < 0.000$). Where strong support with a coefficient of 0.329 shows that green behaviors directly increase eWOM communication. The *H2* relationship has a very high coefficient of 0.713, indicating that green behaviors strongly encourage information adoption. Next, *H3* has the strongest relationship with a coefficient of 0.809, emphasizing the importance of credibility in green behavior. The effect of GBI (*H4*) on CSR was significantly supported, with values of $\beta = 0.316$, $t = 4.513$ and $p < 0.000$. The current direct path (*H4*) is supported with a coefficient of 0.316, indicating that green behaviors alone can enhance CSR perceptions. Hence, the elements of eWOM, such as eWC (*H5*: $\beta = 0.320$, $t = 7.579$, $p < 0.000$), eWIA (*H6* $\beta = 0.114$, $t = 1.960$, $p < 0.050$) and eWSC (*H7*: $\beta = 0.279$, $t = 4.226$, $p < 0.000$), had a significant effect on CSR (see Table 5). Active online discussions about a company's green behaviors positively influence public perceptions of its CSR activities, which shows strong support with a coefficient of 0.320, showing that eWOM communication is a key driver of CSR perception. Furthermore, the *H6* path is weaker (0.114) but still supported, showing a modest link between information adoption and CSR. Strong support with a coefficient of 0.279 for the *H7* path, emphasized the importance of trust and credibility in shaping CSR perceptions. The SEM construct path diagram is presented in Figure 3.

Furthermore, the mediating effects of eWC (*H8*) and eWSC (*H10*) were positive and significantly supported by $\beta = 0.105$, $t = 4.875$, $p < 0.000$ and $\beta = 0.226$, $t = 4.284$, $p < 0.000$, respectively, as presented in Figure 4. Coincidentally, the mediating role of eWIA (*H6* $\beta = 0.086$, $t = 1.932$, $p < 0.069$) was not significantly supported, which indicates that this mediating *n* effect is negligible. Overall the results show that green behaviors indirectly enhance CSR perceptions by encouraging online discussions about these behaviors. Table 5 and Figure 4 present the PLS-SEM results for the proposed mediating hypothesis. Furthermore, the intermediary role of eWOM's indicators exists among GBI and CSR at the 0.105 and 0.226 with positive probability levels, which specifies the involvement of eWC and eWSC in the antecedent of the GBI for CSR. The outcome of eWIA's mediating effect on CSR was insignificant, with a 0.072 probability level. The probability level of 0.072 is considerable but meaningless for this study because of the succeeding $p \leq 0.05$ standards for extreme observations.

Table 5. Structural model results

Hypothetical paths	Path coefficients	SD	*t*-statistics (|O/STDEV|)	Results
H1 = GBI → eWC	0.329***	0.051	6.607	Supported
H2 = GBI → eWIA	0.713***	0.033	21.679	Supported
H3 = GBI → eWSC	0.809***	0.028	29.104	Supported
H4 = GBI → CSR	0.316***	0.07	4.513	Supported
H5 = eWC → CSR	0.320***	0.042	7.579	Supported
H6 = eWIA → CSR	0.114*	0.061	1.96	Supported
H7 = eWSC → CSR	0.279***	0.067	4.226	Supported
H8 = GBI → eWC → CSR	0.105***	0.023	4.875	Supported
H9 = GBI → eWIA → CSR	0.086$^{(p = 0.072)}$	0.044	1.932	Not supported
H10 = RI → eWSC → CSR	0.226***	0.055	4.284	Supported

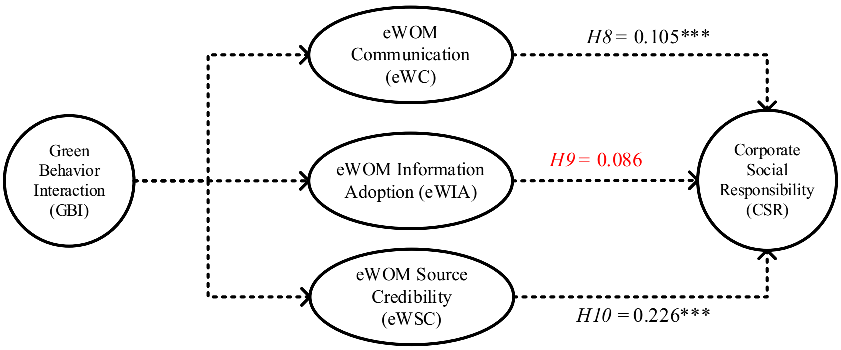
Notes: eWC = eWOM communication; eWIA = eWOM information adoption; eWSC = eWOM source credibility; CSR = corporate social responsibility; GBI = green behavior interactions * = sig. level for <0.05; *** = sig. level for <0.000
Source: Created by authors



Notes: where eWC = eWOM communication; eWIA = eWOM information adoption; eWSC = eWOM source credibility; CSR = corporate social responsibility; GBI = green behavior interactions

Source: Figure by authors

Figure 3. PLS–SEM results



Source: Figure by authors

Figure 4. Structural model result (mediating effect)

4. Discussions

In this study, our objective is to scrutinize the direct impact of GBI on CSR while also assessing the intermediary role played by the three components of eWOM under the lens of the stakeholder theoretical perspective. To provide a robust theoretical foundation for this inquiry, the logistics sector was selected considering eWOM, which amplifies the impact of green logistics practices on stakeholders' green behavior. Ahmed *et al.* (2020) also explained that by incorporating GBIs, the findings contribute to filling gaps in both local and international literature. In the Chinese context, it highlighted that the effectiveness of CSR campaigns that promote green behavior in generating eWOM, eWOM's subfactors that influence consumer perceptions of environmentally focused CSR and successful green initiatives draws actionable lessons. For example, findings highlighted that the impact of GBI on CSR is a subject of interest for researchers and practitioners. For instant, the positive path coefficients of the GBI highlight the importance of interactive information in enhancing CSR and eWOM. This highlight explained that logistics companies could increase interest in implementing green initiatives such as eco-friendly transportation, energy-efficient warehousing and sustainable packaging for the encouragement of environmental consciousness. The analysis of AlSuwaidi *et al.* (2021) showed that CSR initiatives are heightened by green behavior, which helps businesses generate a competitive advantage and build a good image. Achieving a competitive edge is very difficult to win the confidence of buyers and sellers. Our outcome clearly implies that participating in GBI allows businesses to increase eWC, eWIA and eWSC because when individuals engage in GBI, such as when they share detailed product information, eWOM trends are more credible and trustworthy. For this reason, people trust advice with evidence. Furthermore, our results for the GBI indicate that access and interactive information, reviews and comparisons on social sites help decision-makers make informed decisions about how to align products with respect to needs and preferences. Similarly, the greater effect of GBI on eWIA shows that engagement from consumers and businesses is very high, which leads to wider dissemination of eWOM, as more people read, comment and share information, increasing the impact. In addition, as an assurance of our findings, the positive impact of GBI on CSR demonstrates that a business can mitigate CSR-related risk by openly addressing challenges and providing updates on corrective actions, companies demonstrate their commitment to addressing issues promptly toward green behavior. The benefit of positive outcomes for a firm is a good assessment of firm value in terms of social responsibility. Because information interactions may inspire green innovations, green innovation plays a stronger role in promoting the CSR value of a firm. This study provides positive results for promoting transparency, educating stakeholders, raising awareness, enhancing reputation, fostering stakeholder engagement, motivating others, mitigating risks, providing a competitive advantage, ensuring regulatory compliance and supporting the long-term sustainability of CSR initiatives. In summary, internationally, the findings from the current analysis offer insights for other countries grappling with sustainability challenges. For example, China's rapid adoption of green technology and its integration with CSR strategies serve as a model for emerging markets. China's approach to green behavior in CSR with that of other countries, offering a global perspective on best practices and areas for improvement.

With respect to the effect of eWOM indicators on CSR hand, the SEM results reveal a positive influence that helps businesses amplify CSR initiatives by spreading information about their social and environmental responsibilities to a broader audience. Moreover, the findings were similar to the study of Ahmed *et al.* (2020) and Czarnecka *et al.* (2022) concluded a positive relationship between CSR, eWOM and green behavior. This means that positive stories or experiences related to a company's CSR activities on social media or review platforms increase the reach and visibility of CSR efforts for green behavior. The results for eWC reveal that eWOM establishes a constant line of communication between a business and its clients. Especially in the logistics sector of China, platforms like LinkedIn, WeChat and forums dedicated to supply chain discussions can serve as channels for generating eWOM in the logistics sector. Logistics companies that demonstrate genuine commitment to green practices are considered credible sources of sustainable innovation. Stakeholders adopt ideas from successful green initiatives publicized online. A sense of community and shared responsibility can be fostered when businesses constantly update their audience on new CSR projects and achievements. Furthermore, for the eWSC results, this work signifies that eWOM is currently a more trustworthy source than other forms of marketing and corporate communication believed by individuals. A flawless inference about this outcome is that trust in a firm's CSR initiatives can increase when consumers or stakeholders speak openly about their own experiences with the company. Furthermore, with respect to the SEM outcomes for *H8* and *H10*, we conclude that eWC and eWSC transmit information, opinions and recommendations because customers exchange experiences concerning companies' eco-friendly services (e.g. reduced carbon footprint delivery options). eWC and eWSC act as a bridge between the sender (a customer) and the receiver (possible customers). This mediating bridge helps businesses make decisions, manage their business image, increase sales and revenue and build trust. In addition, a high level of monitoring for information adoption on social platforms is relatively difficult, highlighting the significance of the hypothesized relationship, *H9*. The logistic sector was perfectly suited to investigate the proposed research framework for sustainable logistics solutions local and international forms. The interplay between GBIs, eWOM and CSR offers competitive benefits while fostering environmental and social responsibility, building long-term trust and loyalty among stakeholders.

4.1 Theoretical implications

Theoretically, the SEM results confirm an organizational phenomenon that improves performance. This study examines the logistics industry's GBI and social media's function as a mediator. eWOM, particularly its subfactor communication, information adoption and source credibility, plays a critical role in shaping CSR outcomes. The study contributes to ST by investigating how GBIs influence CSR, which is mediated by these eWOM components.

The findings have major theoretical implications for ST, GBIs, CSR and eWOM. For example, first, current research supports ST by showing how consumers, employees and the community impact and are influenced by GBIs and CSR activities. The findings suggest that stakeholder communication through eWOM acts as a strategic tool, reinforcing ST by showing how these interactions shape the perception of a firm's CSR. Second, this study introduces eWOM as a central component in the CSR literature by demonstrating the mediating role of eWOM in the relationship between GBIs and CSR. CSR has traditionally been explored in terms of internal company policies and direct customer contacts, but this research places eWOM as a crucial strategy to increase a firm's legitimacy. Third, we provide an additional contribution to the growing literature on green behavior by examining how interactions around green behaviors on social platforms impact CSR. This finding shows that green enterprises may use social media to promote their practices, tying consumer participation to firm-level CSR achievements. Finally, the current study theoretically positions eWOM as a tool for reducing information asymmetry between firms and stakeholders. This is because information asymmetry often leads to distrust or skepticism in corporate green claims.

Moreover, our research provides innovative insights maintained by empirical data on the relationships among the GBI, eWC, eWSC, eWIA and CSR. Scholars in the CSR sphere have not yet investigated how GBI and eWOM (eWC, eWSC and eWIA) indicators contribute to green behavior and CSR initiatives. To contribute to ST, this study supports the hypothesis that lawmakers or decision-makers can improve corporate performance. This research also shows that the GBI is essential for green logistics growth. In particular, findings reveal that enhancing a firm's CSR and green behavior entails the full administration of GBI on social platforms with the full mediation of eWOM, and organizations may obtain a competitive edge by combining GBI with eWOM. Finally, the collected empirical data reveal that green behavior and CSR initiatives significantly improved through the GBI and eWOM indicators.

4.2 Practical implications

This study has practical implications for policymakers, such as managers, suppliers and consumers, who could benefit from the logistic sector's green behavior and CSR by improving online business policies. First, policymakers could incentivize firms to improve transparency and communication about their green practices on social media and other online platforms. This study proves enhancing the credibility of eWOM is crucial for companies to build trust and influence consumer perceptions effectively. By following current analysis, companies can enhance the credibility of eWOM. Credible eWOM is a trust-builder and a significant driver of brand loyalty and long-term business success. Governments and regulatory bodies can even set guidelines to promote honest, transparent and credible green claims, reducing the risks of greenwashing. Furthermore, this study suggested that companies should also highlight endorsements from credible sources such as industry experts and certifications and leverage employees as advocates to share authentic experiences. However, businesses may actively promote their green behavior and CSR activities via eWOM. By encouraging stakeholders to discuss a firm's green practices online, firms may increase their CSR activities. Second, eWSC plays a significant role in shaping stakeholders' perceptions of a firm's green behavior. To enhance eWSC, firms should partner with trusted sources, such as environmental organizations or influencers, to endorse their sustainability efforts. Furthermore, policymakers could mandate disclosure practices or reporting frameworks that require firms to provide proof of their green initiatives. The subsequent development of industry-wide certifications or third-party verification systems could help improve the eWSC and reduce skepticism around green claims. In conclusion, managers and policymakers must manage eWOM efforts to shape the impact of GBI's impact on CSR. This study presents a fresh paradigm and clear company strategies to increase CSR performance and stakeholder confidence.

These policies provide concrete steps for how logistic companies might best serve their communities by prioritizing online business. More specifically, management should promote their companies' CSR and green behavior by consistently prioritizing eWOM activities. Thus, the managers and owners of firms must concentrate on services for the transportation of goods, which should promote green logistics while operating and manufacturing services in accordance with internationally recognized norms. The findings also revealed that GBI has a significant effect on CSR, and eWOM's indicators play a decisive intermediating role between GBI and CSR, leading scholars to conclude that acceptance of eWOM is key to promoting CSR and GBI for a firm. Therefore, this research recommends maintaining eWOM's vitality through financing and educational programs. In addition, research also recommends that, to enhance CSR performance and green behavior, policymakers should establish transparent policies that logistics firms have implemented. Finally, this work suggests that managers should focus continuously on eWOM and GBI to unlock higher CSR for a long period of time. This can be accomplished by depending on the best possible informative resources that can ensure social responsibility in green logistical services.

5. Conclusion

This study provides evidence that eWOM reduces information asymmetry, enhances credibility, supports informed decisions, improves green consumer behavior, amplifies positive reviews, increases engagement and establishes a feedback loop for the products/services of firms. For CSR, this work offers evidence that GBI positively impacts CSR through high transparency on social platforms, which educates stakeholders, increases awareness, enhances reputation, fosters stakeholder engagement, motivates others, mitigates risks, provides a competitive advantage, ensures regulatory compliance and supports the long-term sustainability of CSR initiatives. As a whole, firms that prioritize the GBI strategy are more likely to benefit from positive eWOM by adapting to green behavior and CSR activities. Companies that actively promote and manage positive eWOM about their CSR initiatives can improve their reputation and achieve good social and environmental changes. The results show that eWOM helps firms align their CSR initiatives with their fundamental beliefs and societal expectations, allowing them to succeed while contributing to society and the environment.

Despite its valuable contributions to scholars, managers, practitioners, and policymakers, this work has several limitations regarding sample foundations, hypothetical relationships, and industrial culture. First, the sample foundation for this work is limited to Chinese logistics firms. The targeted sample may have career generalizability issues because many other companies are connected with the logistical network of China. Second, this research investigates the mediating role of eWOM indicators among GBI and CSR. For future research, we suggest that research could take the eWOM concept as a moderator for GBI and CSR, which could reveal how firms can take up CSR initiatives and GBI administration. Third, our study is limited to one industry of a particular economy; for better verification, scholars may collect data from other industries connected with social platforms. Finally, the outcomes of this work are limited to Chinese industrial culture. This may pose a culture limitation to the results, and future research may consider a broader context to validate the same research assumptions.

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